

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df=pd.read_csv(r"C:\analyticsproject\cluster\fifa.csv",encoding='cp1252')
```

```
In [3]: df.head()
```

Out[3]:

	ID	Name	Age	Nationality	Overall	Potential	Club	Value	Wage	Pre
0	158023	L. Messi	31	Argentina	94	94	FC Barcelona	110500.0	565	
1	20801	Cristiano Ronaldo	33	Portugal	94	94	Juventus	77000.0	405	
2	190871	Neymar Jr	26	Brazil	92	93	Paris Saint-Germain	118500.0	290	
3	193080	De Gea	27	Spain	91	93	Manchester United	72000.0	260	
4	192985	K. De Bruyne	27	Belgium	91	92	Manchester City	102000.0	355	

```
In [4]: df.tail()
```

Out[4]:

	ID	Name	Age	Nationality	Overall	Potential	Club	Value	Wage	Pre
18202	238813	J. Lundstram	19	England	47	65	Crewe Alexandra	60.0		
18203	243165	N. Christoffersson	19	Sweden	47	63	Trelleborgs FF	60.0		
18204	241638	B. Worman	16	England	47	67	Cambridge United	60.0		
18205	246268	D. Walker-Rice	17	England	47	66	Tranmere Rovers	60.0		
18206	246269	G. Nugent	16	England	46	66	Tranmere Rovers	60.0		

```
In [5]: df.sample(5)
```

Out[5]:

	ID	Name	Age	Nationality	Overall	Potential	Club	Value	Wage
4578	242380	Stephen Eustáquio	21	Portugal	71	83	GD Chaves	3800.0	5
16779	237653	C. Rojas	20	Colombia	56	65	Envigado FC	140.0	1
4127	240690	N. González	20	Argentina	71	83	VfB Stuttgart	4300.0	17
358	201454	F. Ghoulam	27	Algeria	81	82	Napoli	16500.0	68
7441	216271	J. van Aken	24	Netherlands	68	74	Sheffield Wednesday	1000.0	9


In [6]: `df.shape`

Out[6]: (18207, 18)

In [7]: `df.columns`

Out[7]: Index(['ID', 'Name', 'Age', 'Nationality', 'Overall', 'Potential', 'Club', 'Value', 'Wage', 'Preferred Foot', 'International Reputation', 'Skill Moves', 'Position', 'Joined', 'Contract Valid Until', 'Height', 'Weight', 'Release Clause'], dtype='object')

In [8]: `df.isnull().sum()`

```
Out[8]: ID                0
Name                0
Age                0
Nationality        0
Overall            0
Potential          0
Club              241
Value             252
Wage              0
Preferred Foot     0
International Reputation  48
Skill Moves       48
Position          0
Joined            0
Contract Valid Until 289
Height            0
Weight            0
Release Clause     0
dtype: int64
```

In [9]: `df.dtypes`

```
Out[9]: ID                int64
        Name              object
        Age               int64
        Nationality       object
        Overall            int64
        Potential         int64
        Club              object
        Value             float64
        Wage              int64
        Preferred Foot     object
        International Reputation float64
        Skill Moves       float64
        Position          object
        Joined            int64
        Contract Valid Until object
        Height            float64
        Weight            float64
        Release Clause     float64
        dtype: object
```

```
In [10]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18207 entries, 0 to 18206
Data columns (total 18 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   ID                    18207 non-null int64
 1   Name                  18207 non-null object
 2   Age                   18207 non-null int64
 3   Nationality           18207 non-null object
 4   Overall                18207 non-null int64
 5   Potential              18207 non-null int64
 6   Club                  17966 non-null object
 7   Value                 17955 non-null float64
 8   Wage                  18207 non-null int64
 9   Preferred Foot        18207 non-null object
10   International Reputation 18159 non-null float64
11   Skill Moves           18159 non-null float64
12   Position              18207 non-null object
13   Joined                18207 non-null int64
14   Contract Valid Until   17918 non-null object
15   Height                18207 non-null float64
16   Weight                18207 non-null float64
17   Release Clause         18207 non-null float64
dtypes: float64(6), int64(6), object(6)
memory usage: 2.5+ MB
```

```
In [11]: df.describe()
```

Out[11]:

	ID	Age	Overall	Potential	Value	Wa
count	18207.000000	18207.000000	18207.000000	18207.000000	17955.000000	18207.0000
mean	214298.338606	25.122206	66.238699	71.307299	2444.530214	9.7313
std	29965.244204	4.669943	6.908930	6.136496	5626.715434	21.9992
min	16.000000	16.000000	46.000000	48.000000	10.000000	0.0000
25%	200315.500000	21.000000	62.000000	67.000000	325.000000	1.0000
50%	221759.000000	25.000000	66.000000	71.000000	700.000000	3.0000
75%	236529.500000	28.000000	71.000000	75.000000	2100.000000	9.0000
max	246620.000000	45.000000	94.000000	95.000000	118500.000000	565.0000

```
In [12]: df['Club']=df['Club'].fillna('No Club')
```

```
In [13]: a=df['Value'].median()
df['Value']=df['Value'].fillna(a)
```

```
In [14]: c=df['Skill Moves'].mode()[0]
df['Skill Moves']=df['Skill Moves'].fillna(c)
```

```
In [15]: df['Contract Valid Until']=pd.to_datetime(df['Contract Valid Until'])
```

```
In [16]: df=df.dropna(subset=['Contract Valid Until'])
```

```
In [17]: df.isnull().sum()
```

```
Out[17]: ID          0
Name          0
Age          0
Nationality   0
Overall       0
Potential     0
Club          0
Value         0
Wage         0
Preferred Foot 0
International Reputation 0
Skill Moves   0
Position      0
Joined        0
Contract Valid Until 0
Height        0
Weight        0
Release Clause 0
dtype: int64
```

```
In [18]: df.duplicated().sum()
```

```
Out[18]: np.int64(0)
```

```
In [19]: print(df['Age'].quantile(0.25))
print(df['Age'].quantile(0.50))
print(df['Age'].quantile(0.75))
```

```
21.0
```

```
25.0
```

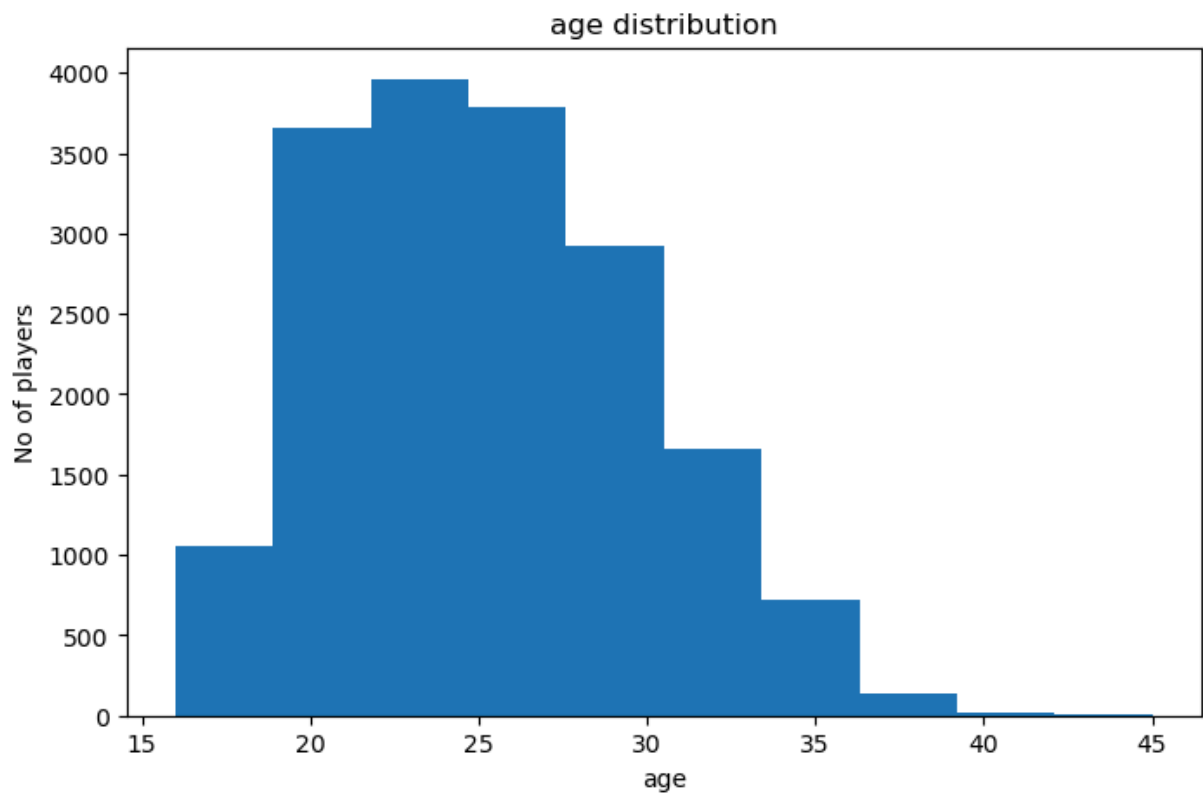
```
28.0
```

```
In [20]: df['Age'].describe()
```

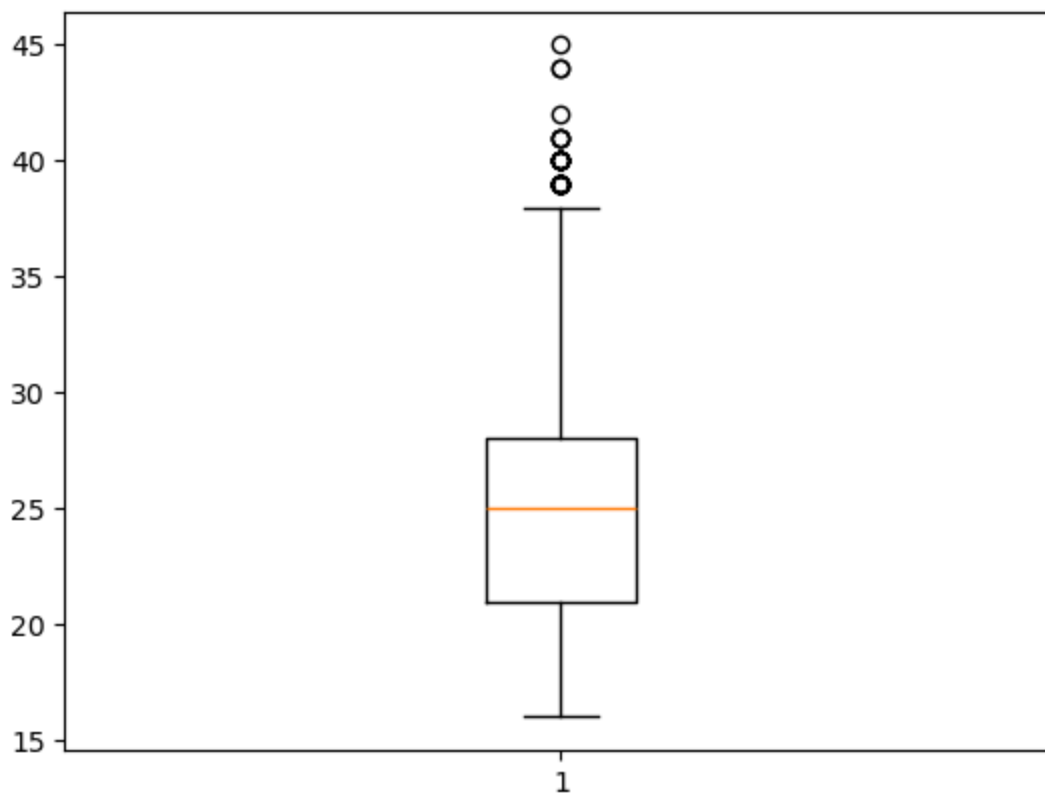
```
Out[20]: count    17918.000000
mean         25.105257
std           4.675372
min          16.000000
25%          21.000000
50%          25.000000
75%          28.000000
max          45.000000
Name: Age, dtype: float64
```

```
In [21]: import matplotlib.pyplot as plt
```

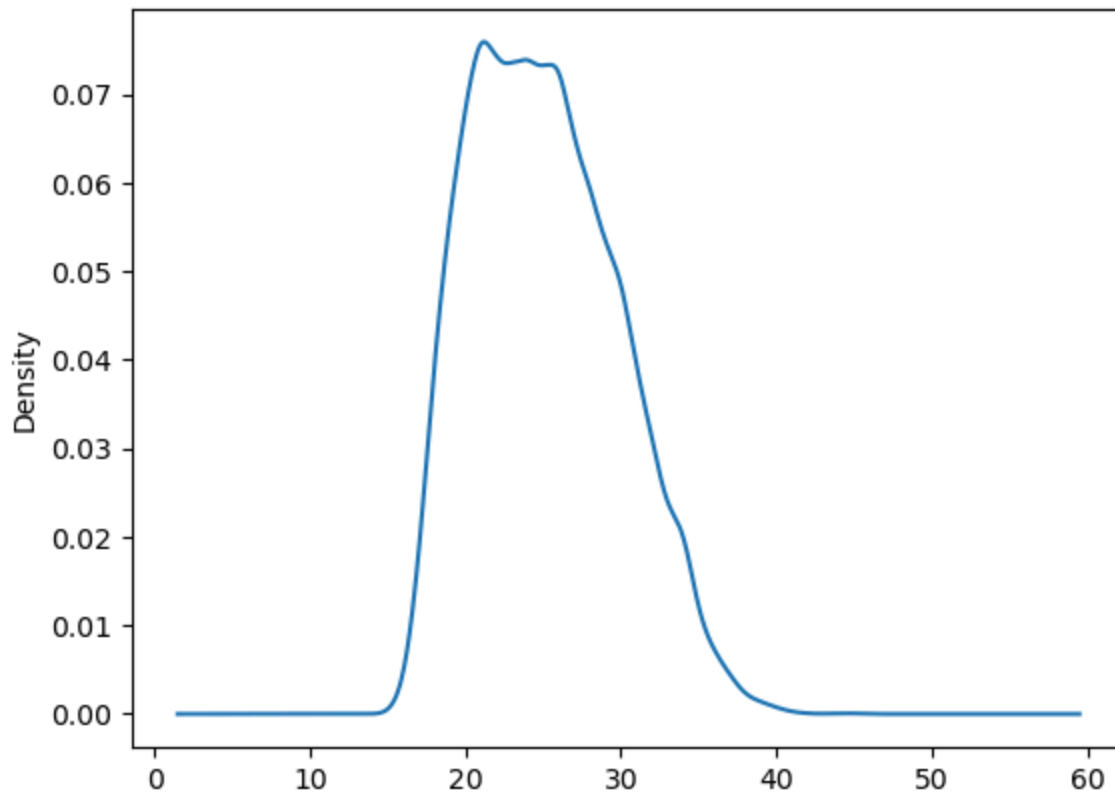
```
In [22]: plt.figure(figsize=(8,5))
plt.hist(df['Age'])
plt.title('age distribution')
plt.xlabel('age')
plt.ylabel('No of players')
plt.show()
```



```
In [23]: plt.boxplot(df['Age'])  
plt.show()
```



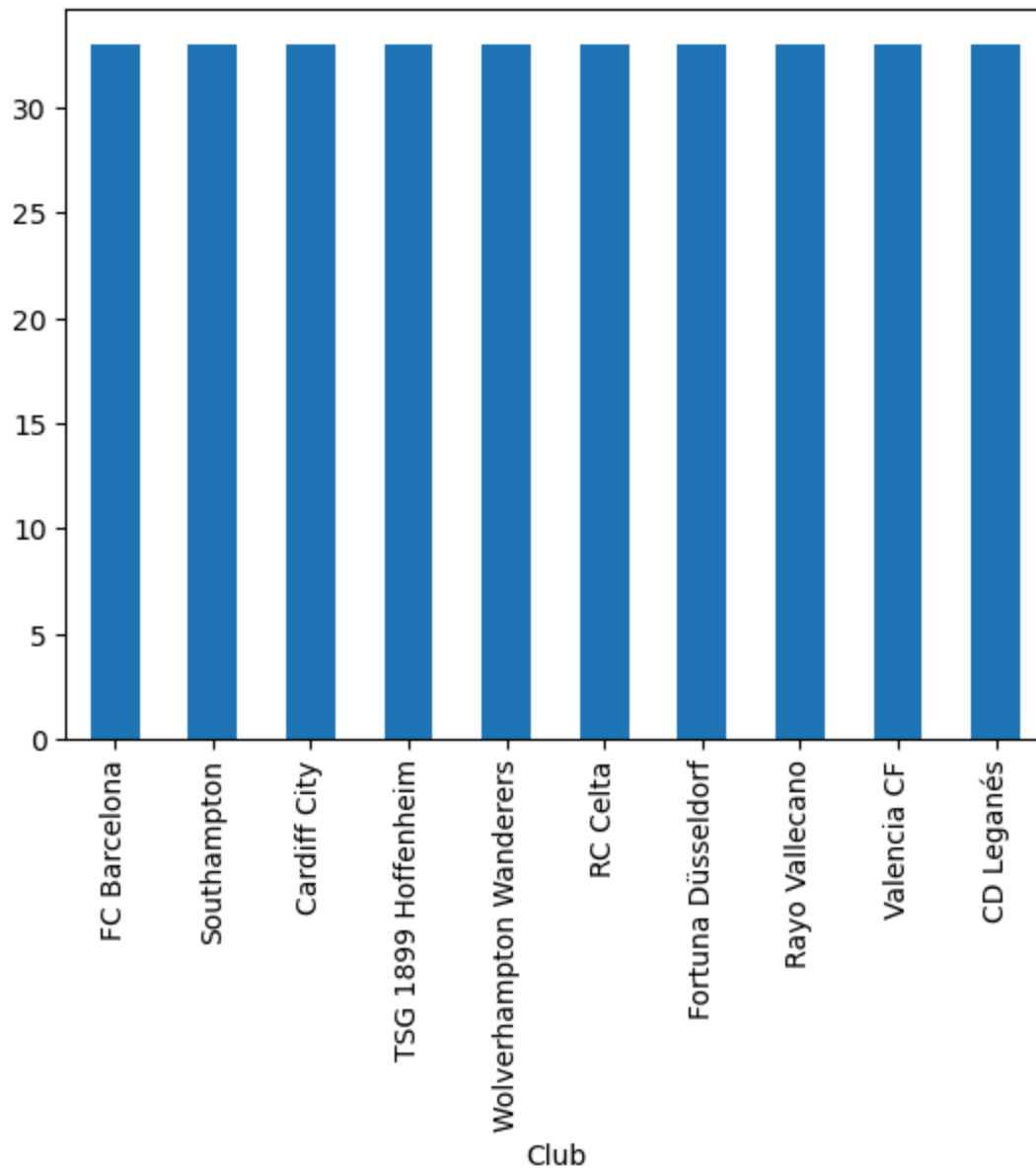
```
In [24]: df['Age'].plot(kind='density')  
plt.show()
```



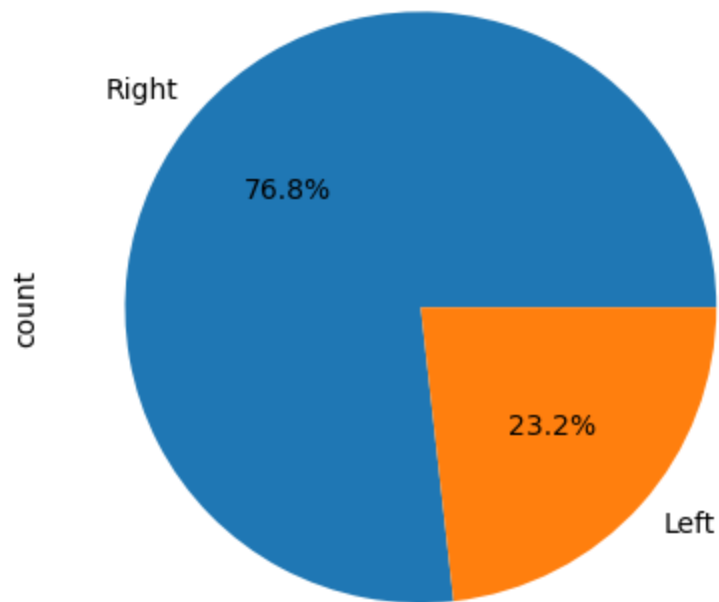
```
In [25]: df['Club'].value_counts()
```

```
Out[25]: Club
FC Barcelona          33
Southampton           33
Cardiff City          33
TSG 1899 Hoffenheim   33
Wolverhampton Wanderers 33
..
Vitória               20
Paraná                20
Sligo Rovers          19
Limerick FC           19
Derry City            18
Name: count, Length: 651, dtype: int64
```

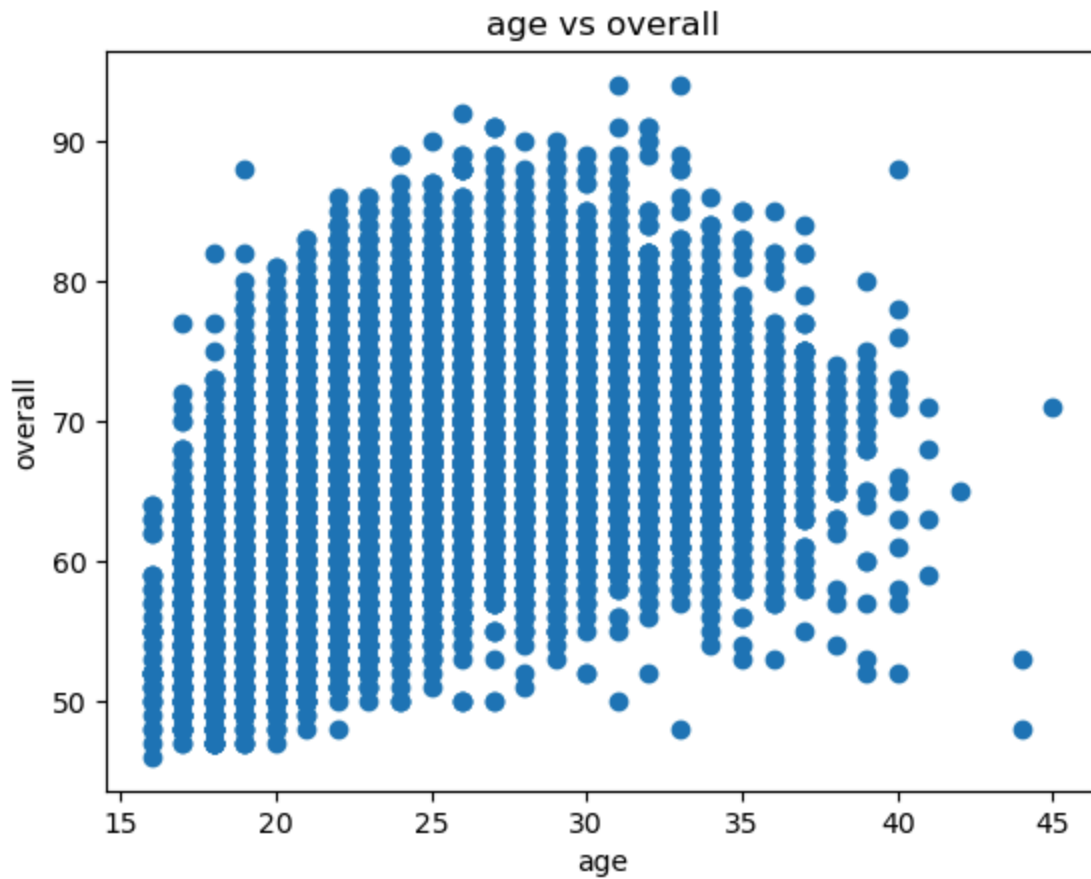
```
In [26]: df['Club'].value_counts().head(10).plot(kind='bar')
plt.show()
```



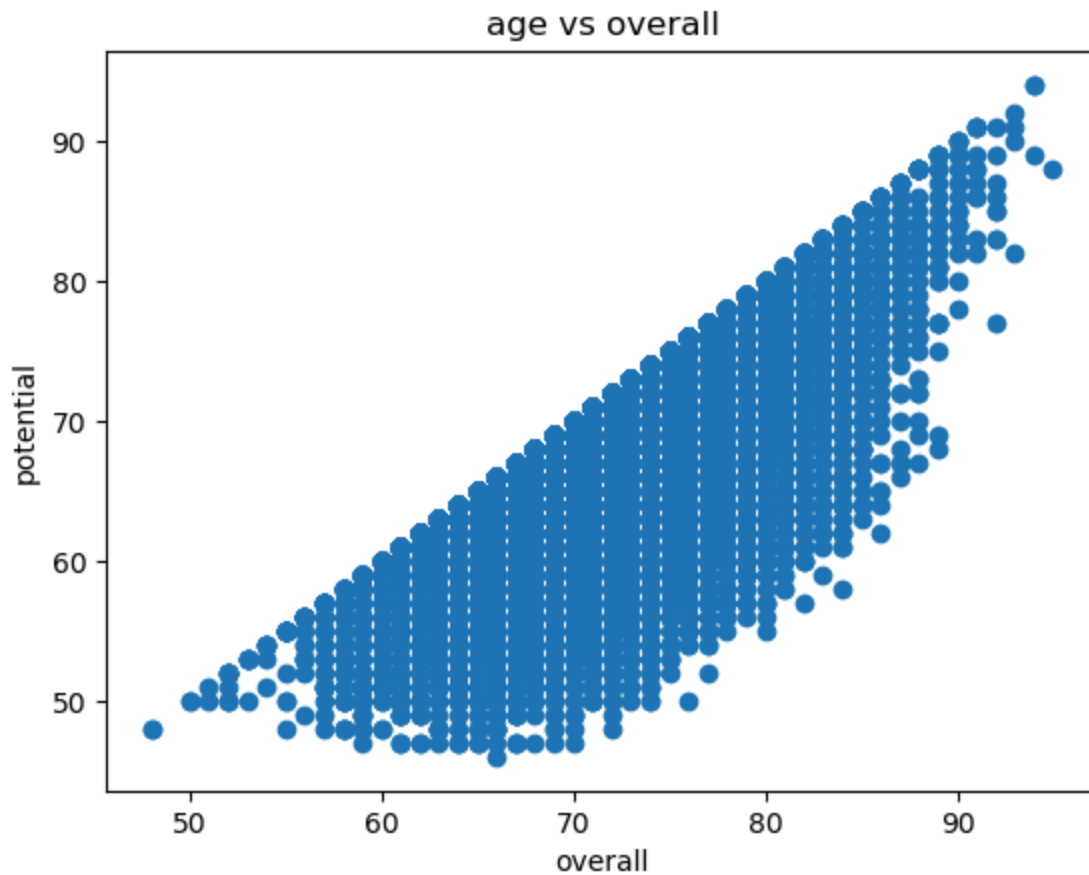
```
In [27]: df['Preferred Foot'].value_counts().plot(kind='pie', autopct='%1.1f%%')  
plt.show()
```

```
In [28]: plt.scatter(df['Age'],df['Overall'])  
plt.title('age vs overall')  
plt.xlabel('age')  
plt.ylabel('overall')  
plt.show()
```



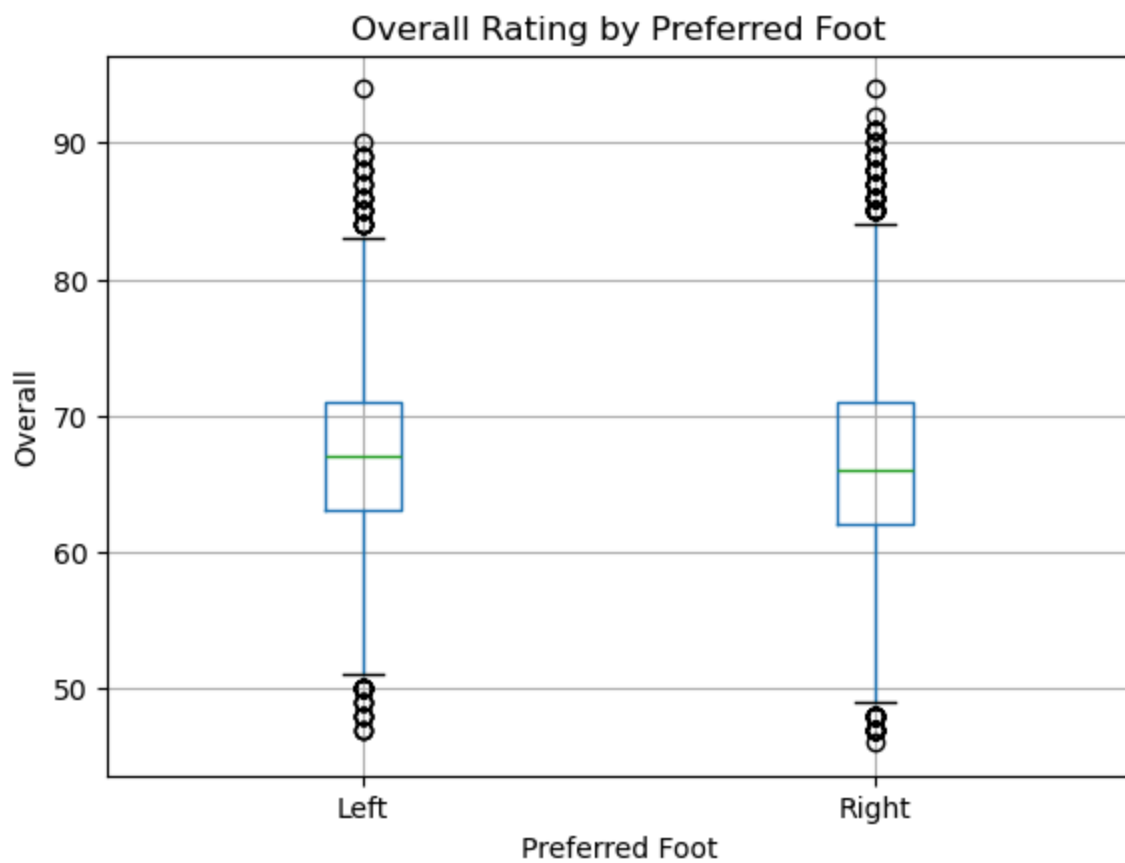
```
In [29]: plt.scatter(df['Potential'],df['Overall'])  
plt.title('age vs overall')  
plt.xlabel('overall')  
plt.ylabel('potential')  
plt.show()
```



```
In [30]: df['Age'].corr(df['Overall'])
```

```
Out[30]: np.float64(0.45325165498967435)
```

```
In [31]: df.boxplot(column="Overall", by="Preferred Foot")
plt.title("Overall Rating by Preferred Foot")
plt.suptitle("")
plt.xlabel("Preferred Foot")
plt.ylabel("Overall")
plt.show()
```



```
In [32]: data=df.groupby('Position')['Overall'].mean()
```

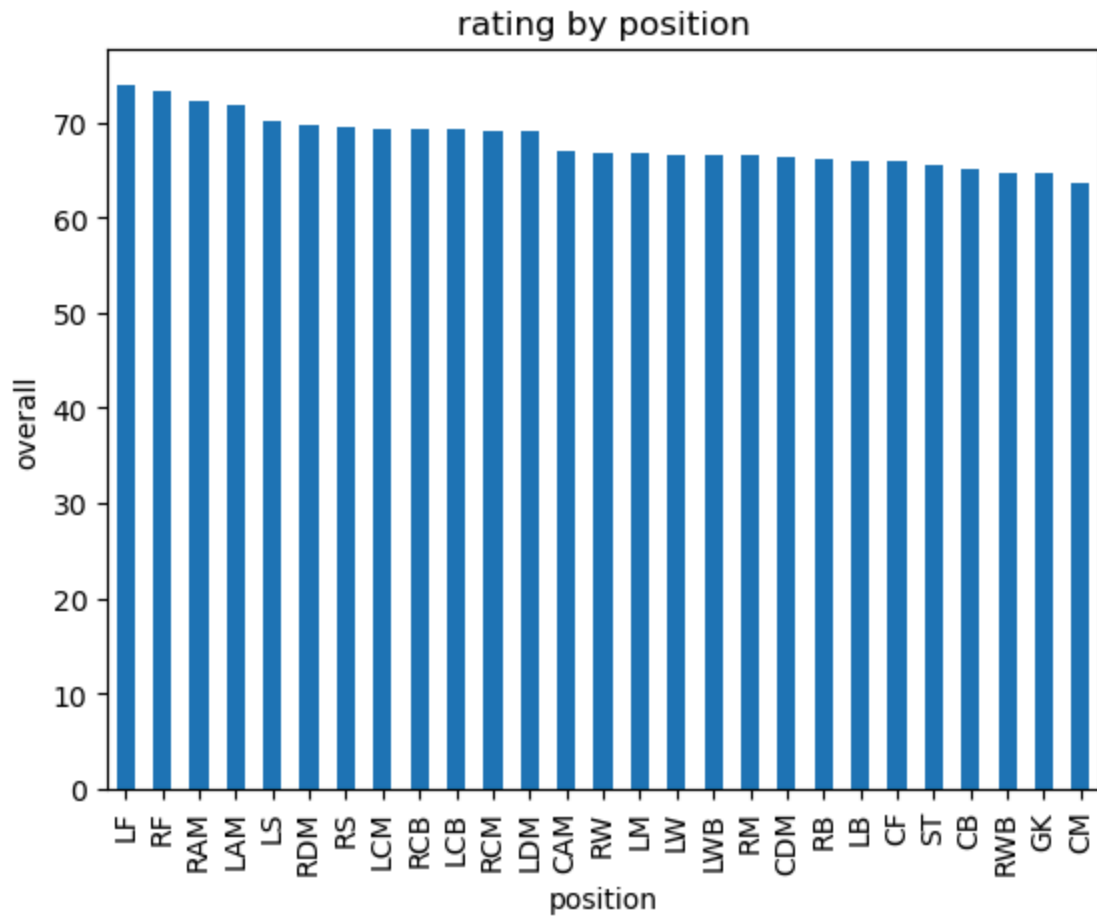
```
In [33]: data
```

Out[33]: Position

CAM	66.877637
CB	64.992018
CDM	66.313034
CF	65.824324
CM	63.697168
GK	64.562751
LAM	71.857143
LB	65.846743
LCB	69.202512
LCM	69.334190
LDM	69.100418
LF	73.866667
LM	66.675875
LS	70.077670
LW	66.639037
LWB	66.512821
RAM	72.285714
RB	66.032334
RCB	69.315951
RCM	69.105943
RDM	69.739837
RF	73.250000
RM	66.473070
RS	69.537313
RW	66.835616
RWB	64.655172
ST	65.512676

Name: Overall, dtype: float64

```
In [34]: data.sort_values(ascending=False).plot(kind='bar')
plt.title('rating by position')
plt.xlabel('position')
plt.ylabel('overall')
plt.show()
```



```
In [35]: df.corr(numeric_only=True)
```

Out[35]:

	ID	Age	Overall	Potential	Value	Wage	International Reputation
ID	1.000000	-0.740059	-0.417698	0.047481	-0.138427	-0.204988	-0.355581
Age	-0.740059	1.000000	0.453252	-0.252799	0.077533	0.143944	0.254064
Overall	-0.417698	0.453252	1.000000	0.660850	0.631363	0.575940	0.501142
Potential	0.047481	-0.252799	0.660850	1.000000	0.579379	0.488999	0.374155
Value	-0.138427	0.077533	0.631363	0.579379	1.000000	0.858032	0.655271
Wage	-0.204988	0.143944	0.575940	0.488999	0.858032	1.000000	0.670736
International Reputation	-0.355581	0.254064	0.501142	0.374155	0.655271	0.670736	1.000000
Skill Moves	-0.057347	0.028737	0.415865	0.354908	0.317185	0.264344	0.208307
Joined	0.208455	-0.203358	-0.170142	-0.048755	-0.115502	-0.144047	-0.133809
Height	-0.088962	0.082072	0.038241	-0.010101	0.002768	0.019836	0.034891
Weight	-0.190523	0.229824	0.154095	-0.007664	0.046600	0.065545	0.088427
Release Clause	-0.121957	0.059075	0.600857	0.566161	0.973462	0.829404	0.622710

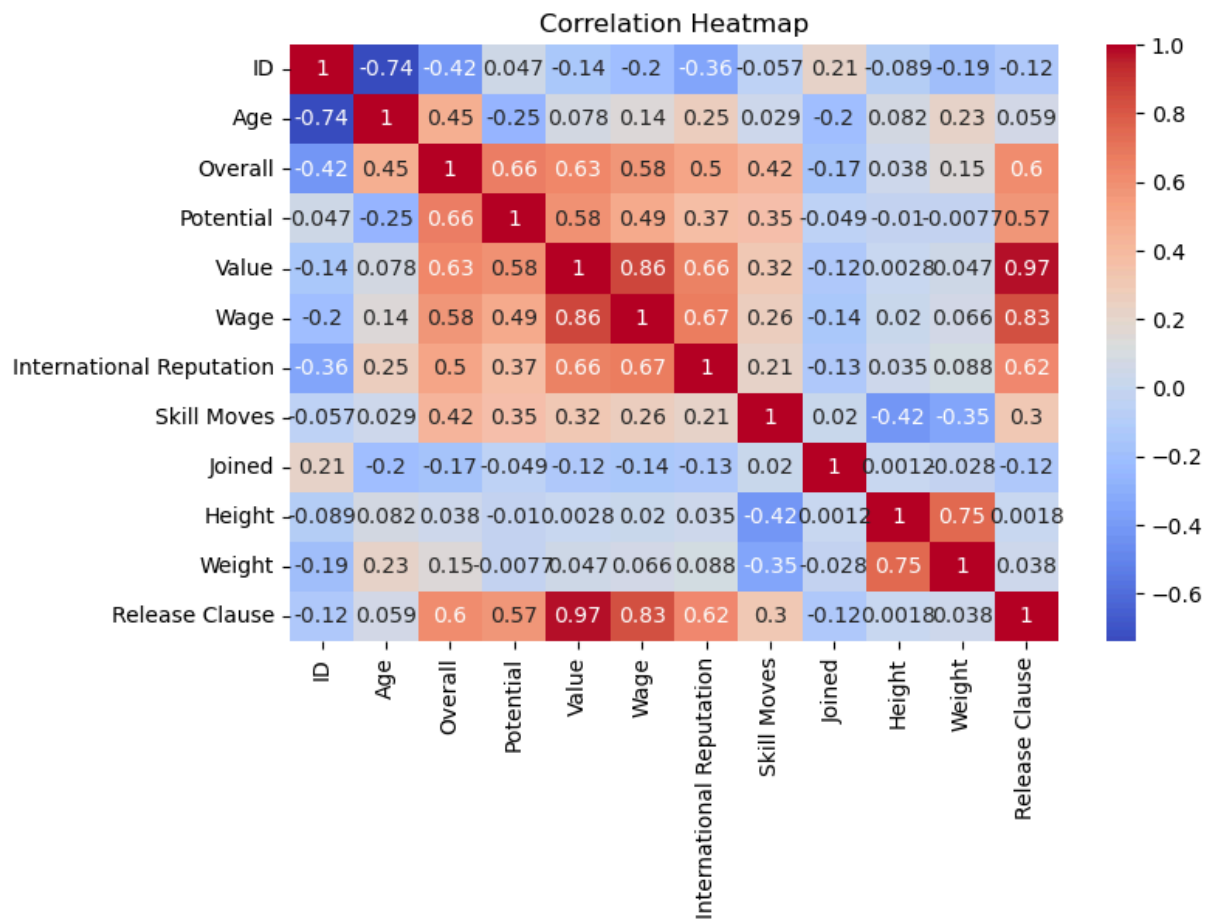
In [36]: `import seaborn as sns`

```

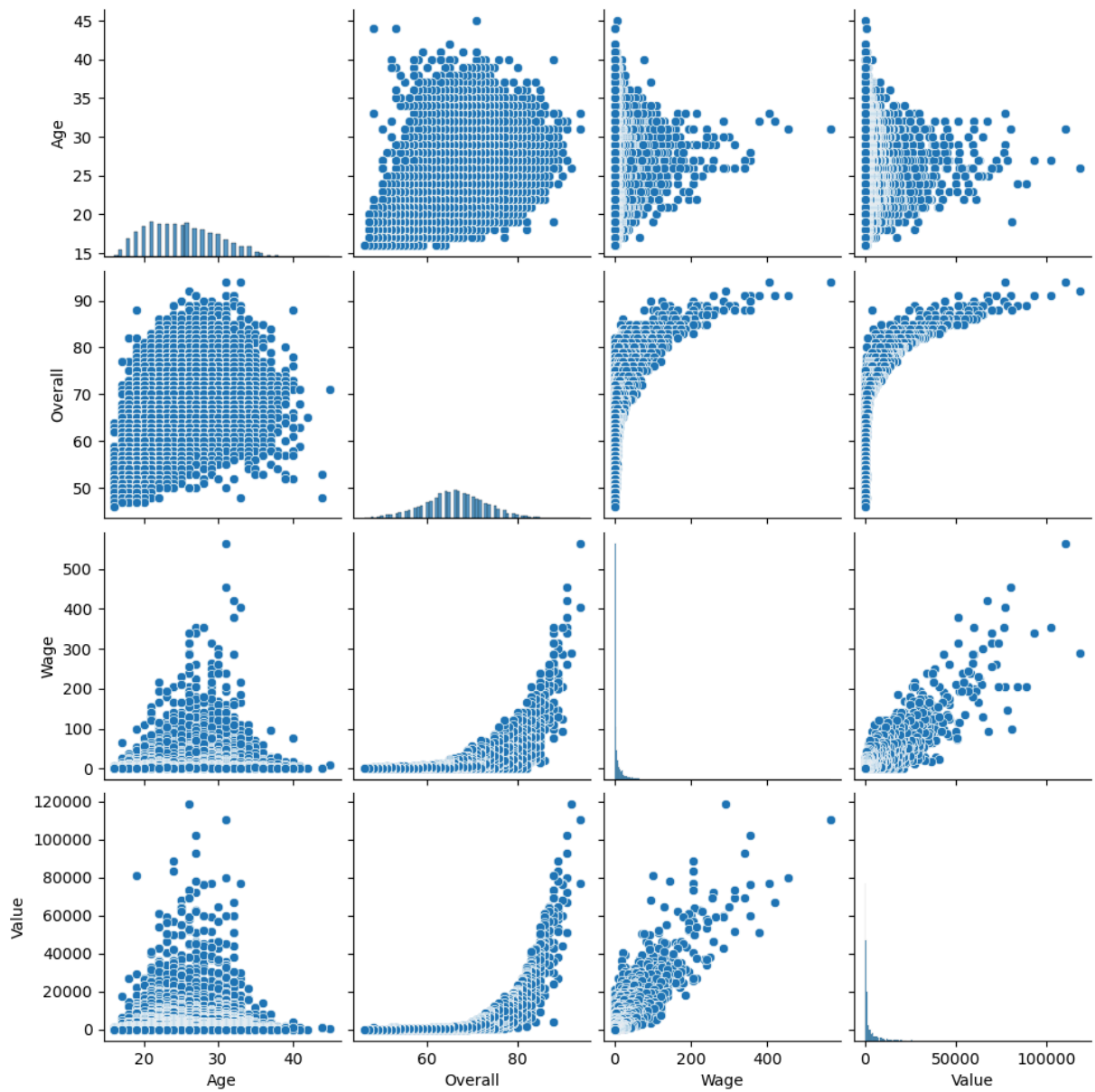
In [37]: plt.figure(figsize=(8,5))
sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap="coolwarm")

plt.title("Correlation Heatmap")
plt.show()

```



```
In [38]: sns.pairplot(df[["Age", "Overall", "Wage", "Value"]])
plt.show()
```

```
In [39]: pd.pivot_table(
    df,
    values="Wage",
    index="Position",
    aggfunc="mean"
)
```

Out[39]:

Wage	
Position	
CAM	10.337553
CB	7.810148
CDM	9.434829
CF	10.216216
CM	8.443718
GK	6.916667
LAM	26.142857
LB	8.840613
LCB	11.697017
LCM	14.349614
LDM	12.058577
LF	44.666667
LM	9.736648
LS	15.334951
LW	13.312834
LWB	9.076923
RAM	19.095238
RB	8.760252
RCB	12.883436
RCM	14.552972
RDM	12.247967
RF	52.687500
RM	9.626571
RS	14.522388
RW	14.630137
RWB	8.597701
ST	10.040845

```
In [40]: df.groupby(  
    ["Position", "Preferred Foot"]  
)[ "Overall" ].mean()
```

```

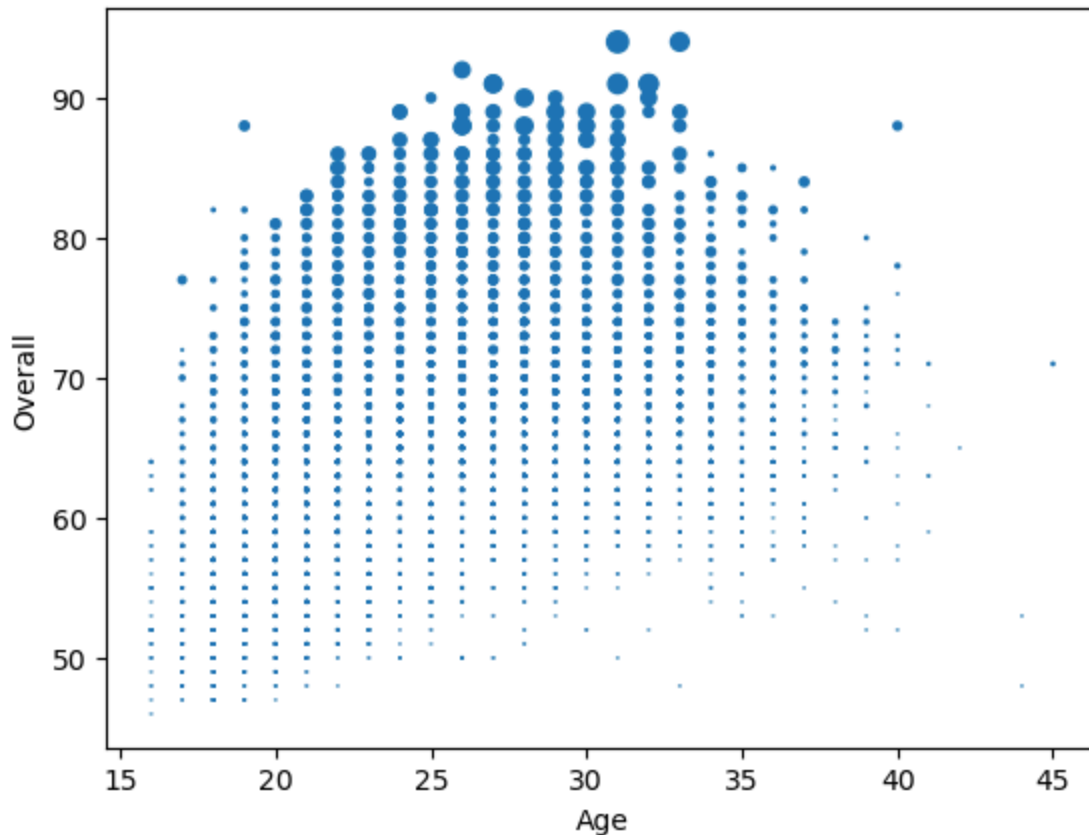
Out[40]: Position Preferred Foot
CAM      Left      68.162055
          Right     66.410072
CB        Left      65.704142
          Right     64.822034
CDM       Left      66.177419
          Right     66.333744
CF        Left      65.666667
          Right     65.875000
CM        Left      65.380567
          Right     63.329204
GK        Left      66.307317
          Right     64.362619
LAM       Left      75.111111
          Right     69.416667
LB        Left      66.036617
          Right     64.468354
LCB       Left      69.022901
          Right     69.328000
LCM       Left      70.288660
          Right     69.017123
LDM       Left      68.183673
          Right     69.336842
LF        Left      76.333333
          Right     73.250000
LM        Left      65.248768
          Right     67.527941
LS        Left      70.575758
          Right     69.982659
LW        Left      65.314286
          Right     67.156134
LWB       Left      66.166667
          Right     68.416667
RAM       Left      72.444444
          Right     72.166667
RB        Left      63.642857
          Right     66.059011
RCB       Left      67.906250
          Right     69.388710
RCM       Left      70.780488
          Right     68.907514
RDM       Left      69.304348
          Right     69.784753
RF        Left      74.666667
          Right     72.400000
RM        Left      68.662651
          Right     65.842775
RS        Left      68.837838
          Right     69.695122
RW        Left      70.063158
          Right     65.700000
RWB       Left      63.500000
          Right     64.710843
ST        Left      66.372414
          Right     65.377174

```

Name: Overall, dtype: float64

```
In [41]: plt.scatter(
    df["Age"],
    df["Overall"],
    s=df["Wage"]/10
)

plt.xlabel("Age")
plt.ylabel("Overall")
plt.show()
```



```
In [42]: df["Contract Duration"] = (
    df["Contract Valid Until"].dt.year - df["Joined"]
)
```

```
In [43]: df["Potential Growth"] = df["Potential"] - df["Overall"]
```

```
In [44]: bins = [15,20,25,30,50]
labels = ["Young", "Early Career", "Peak", "Veteran"]

df["Age Group"] = pd.cut(
    df["Age"],
    bins=bins,
    labels=labels
)
```

```
In [45]: df["Wage_M"] = df["Wage"] / 1000000
```

```
In [46]: df["Value/Wage"] = df["Value"] / df["Wage"]
```

```
In [47]: def rating(x):
    if x >= 90:
        return "Elite"
    elif x >= 80:
        return "Excellent"
    elif x >= 70:
        return "Good"
    else:
        return "Average"

df["Rating Category"] = df["Overall"].apply(rating)
```

```
In [48]: df.shape
```

```
Out[48]: (17918, 24)
```

```
In [49]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 17918 entries, 0 to 18206
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                    17918 non-null  int64
1   Name                                17918 non-null  object
2   Age                                  17918 non-null  int64
3   Nationality                         17918 non-null  object
4   Overall                             17918 non-null  int64
5   Potential                           17918 non-null  int64
6   Club                                17918 non-null  object
7   Value                               17918 non-null  float64
8   Wage                                17918 non-null  int64
9   Preferred Foot                       17918 non-null  object
10  International Reputation              17918 non-null  float64
11  Skill Moves                          17918 non-null  float64
12  Position                             17918 non-null  object
13  Joined                               17918 non-null  int64
14  Contract Valid Until                 17918 non-null  datetime64[ns]
15  Height                              17918 non-null  float64
16  Weight                              17918 non-null  float64
17  Release Clause                      17918 non-null  float64
18  Contract Duration                   17918 non-null  int64
19  Potential Growth                    17918 non-null  int64
20  Age Group                           17918 non-null  category
21  Wage_M                              17918 non-null  float64
22  Value/Wage                          17918 non-null  float64
23  Rating Category                     17918 non-null  object
dtypes: category(1), datetime64[ns](1), float64(8), int64(8), object(6)
memory usage: 3.3+ MB
```

```
In [50]: df.head()
```

Out[50]:

	ID	Name	Age	Nationality	Overall	Potential	Club	Value	Wage	Pre
0	158023	L. Messi	31	Argentina	94	94	FC Barcelona	110500.0	565	
1	20801	Cristiano Ronaldo	33	Portugal	94	94	Juventus	77000.0	405	
2	190871	Neymar Jr	26	Brazil	92	93	Paris Saint-Germain	118500.0	290	
3	193080	De Gea	27	Spain	91	93	Manchester United	72000.0	260	
4	192985	K. De Bruyne	27	Belgium	91	92	Manchester City	102000.0	355	

5 rows × 24 columns

